

THE EXACT MAXIMUM GROWTH FACTOR FOR COMPLETE PIVOTING ON REAL 5×5 MATRICES

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ABSTRACT. Let $\rho_5^{\mathbb{R}}$ be the largest element growth possible in exact Gaussian elimination with complete pivoting on real 5×5 matrices, with every legal choice at a pivot tie allowed. We prove

$$\rho_5^{\mathbb{R}} = \alpha = 4.132517078632472854223346853277 \dots,$$

where α is the algebraic lower bound identified by Chen, Edelman, and Urschel, specified by their degree-61 polynomial and a rational isolating interval. Our contribution is the global upper bound. An exact two-step inverse-elimination representation retains all initial and intermediate entry bounds. A constructive diagonal contraction sends every positive final-pivot value to one of two saturated tail geometries, without changing that value. One geometry admits a complete certified bound; in the remaining geometry, three feasible paths eliminate six variables and produce a seventeen-variable common framework with an attained fibre maximum. Finite rational certificates cover that framework. The terminal rules distinguish infeasibility above a rational threshold from safety at the exact algebraic bound. Their final composition binds every remaining source to its original root and has no uncovered region. The proof combines exact rational certificate verification with source-preserving composition of the resulting bounds.

1. THE PROBLEM AND THE RESULT

Complete pivoting selects a largest-modulus entry of the current Schur complement at every step of Gaussian elimination. Although this rule controls local multipliers, successive Schur updates can generate entries larger than those in the input. Sharp growth factors in small dimensions provide concrete tests of what global information is retained, or lost, by general pivot-growth estimates.

The values in dimensions three and four are $9/4$ and 4 , respectively; classical work includes Cryer [4] and Cohen [3]. Numerical evidence for a fifth-pivot value near 4.1325 was reported by Day and Peterson [5] and subsequently reproduced by Gould [6]. Chen, Edelman, and Urschel [2] identified the specific degree-61 algebraic value α and proved a lower bound by a completely pivoted matrix. Their Theorem 2.2 proves optimality within a prescribed equality family and supplies the lower-bound direction used here; their Theorem 3.5 gives the global upper bound 4.84 . Theorem 1.2

establishes the equality in their Conjecture 2.1. The constant, its defining polynomial, and the equality pattern are due to Chen, Edelman, and Urschel; the advance here is the matching global upper bound.

The present result concerns a fixed dimension and the real field. It is separate from estimates as the dimension tends to infinity, including the recent results of Bisain, Edelman, and Urschel [1] and Shah and Urschel [8]. Background on growth and its relation to numerical error can be found in Day and Peterson [5] and Higham and Higham [7].

Definition 1.1. For $0 \neq A \in \mathbb{R}^{n \times n}$, let π denote a legal complete-pivoting path in exact arithmetic. Write $A_\pi^{(j)}$ for the trailing Schur matrix at stage j , before its leading pivot is eliminated. Define

$$\rho(A, \pi) = \frac{\max_j \|A_\pi^{(j)}\|_{\max}}{\|A\|_{\max}}, \quad \rho_n^{\mathbb{R}} = \sup_{A \neq 0} \sup_{\pi \text{ legal}} \rho(A, \pi).$$

If the current pivot is zero, the trailing matrix is zero and elimination terminates; all subsequent pivot magnitudes are taken to be zero.

We normalize $\|A\|_{\max} = 1$. The first pivot magnitude is then one, and complete pivoting gives

$$\rho(A, \pi) = \max_{1 \leq j \leq n} p_j, \quad (1)$$

where p_j is the magnitude of the chosen pivot. Row and column permutations may be absorbed into the original matrix. Thus it suffices to prove the upper bound for every matrix that is completely pivoted in the fixed diagonal order. This does not assume uniqueness of a largest entry, nor equality of the results of different tied paths.

Theorem 1.2. *For exact complete-pivoting Gaussian elimination on real 5×5 matrices, $\rho_5^{\mathbb{R}} = \alpha$. Here α is the root specified in Section 2. The upper bound holds for every legal pivot path, including paths with ties and inputs for which elimination terminates early.*

The proof has two analytic reductions and two computational components. First, the first four pivot magnitudes are at most $4 < \alpha$. Second, every positive fifth-pivot value has a same-value representative whose last 2×2 Schur block lies on a diagonal-saturated or cross-saturated boundary. The cross-saturated domain is covered by Proposition 6.1. The remaining negative diagonal domain is covered by Proposition 9.1, after the exact fibre reduction in Section 7. Section 8 states the certificate semantics and composition argument. The proof does not require a classification of all equality matrices.

2. THE CONSTANT AND AN ATTAINED LOWER BOUND

Let

$$P_{61}(g) = \sum_{i=0}^{61} c_i g^i \quad (2)$$

be the polynomial of Chen, Edelman, and Urschel [2]. Its leading coefficient is 59049 and its constant coefficient is

$$-400768351683901408958416200638054732473753927680.$$

Supplementary Material S1 gives the full ascending coefficient list and exact decimal rational endpoints $L < U$, with $U - L = 10^{-200}$, isolating one real root. Throughout, α means this root and $\alpha_- = L$. There is also a short exact identification with the published constant. All 62 integer coefficients agree with the published polynomial. For

$$Q(t) = (1+t)^{61} P_{61}\left(\frac{4+5t}{1+t}\right),$$

the coefficients of degrees $0, \dots, 8$ are negative and those of degrees $9, \dots, 61$ are positive. Descartes' rule of signs therefore gives at most one positive root of Q , hence at most one root of P_{61} in $(4, 5)$. Exact evaluation gives $P_{61}(L) < 0 < P_{61}(U)$. Consequently α is the unique root in $(4, 5)$. The exact coefficient comparison, transformed coefficients, and endpoint signs are recorded in Supplementary Material S1. For orientation,

$$\begin{aligned} 4.132517078632472854223346853277 &< \alpha, \\ \alpha &< 4.132517078632472854223346853278. \end{aligned}$$

All certificate comparisons use the full rational endpoints, not these truncated decimals. We also use

$$\Lambda = \frac{1653}{400}, \quad \gamma = \frac{4132517}{10^6}, \quad 4 < \Lambda < \gamma < \alpha_- < \alpha. \quad (3)$$

The lower bound is already a theorem of [2, Theorem 2.2]. To identify the constant unambiguously in the computational archive, a supplementary foundational module also reconstructs that candidate independently. It uses integer polynomials P_1, P_2, P_3 in (x, y, z, g) and

$$J = \det \frac{\partial(P_1, P_2, P_3)}{\partial(x, y, z)}.$$

A rational Krawczyk enclosure isolates a unique zero of (P_1, P_2, P_3, J) in an explicit box. The x, y, z radii are 10^{-180} ; the g -interval is $[L, U]$. An exact elimination identity, exclusion of the other resultant factors, and a Sturm count identify $g = \alpha$.

Rational reconstruction at this zero gives

$$A_* = \begin{pmatrix} 1 & 1 & -z/y & -z & z(x+z)/(x+1) \\ x & x+z+1 & b_{23} & -1 & 1 \\ -1 & z & b_{33} & -1 & -1 \\ y & T & 1 & 1 & 1 \\ z & -1 & 1 & -1 & 1 \end{pmatrix}. \quad (4)$$

The exact rational functions T, b_{23}, b_{33} , the defining equations, and the isolating box are archived with the reconstruction code. For orientation,

$$(x, y, z) \approx (-0.6175326768818271, -0.7791507422163111, 0.4532249098468375).$$

These decimal approximations are not substituted into the certificate. All reconstruction denominators are separated from zero. Entry bounds at equality follow from rational identities and the root equations; the other bounds have strictly positive rational interval margins. The pivot magnitudes are

$$1, \quad 1+z, \quad p_{3,*}, \quad \alpha/2, \quad \alpha,$$

with $1.453 < 1+z < 1.454$ and $2.074 < p_{3,*} < 2.075$. In particular, the final 2×2 tail is exactly

$$\frac{\alpha}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

This is an attained lower bound, not a conclusion drawn solely from a polynomial factor or floating-point optimization.

3. AN ELEMENTARY REDUCTION TO THE FIFTH PIVOT

We include the small-dimensional bounds used in all root domains. This also handles earlier-stage growth without an extra hypothesis.

Lemma 3.1. *For a normalized real 3×3 matrix completely pivoted in the indicated order, let its pivot magnitudes be $1, t, g$. Then*

$$0 \leq t \leq 2, \quad g \leq \Phi(t), \quad \Phi(t) = \begin{cases} 2t, & 0 \leq t \leq 1, \\ t(3-t), & 1 \leq t \leq 2. \end{cases}$$

Each branch is sharp. In particular, $g \leq 9/4$.

Proof. The first Schur update is a difference of two quantities of magnitude at most one, so $t \leq 2$. If $t = 0$, its whole trailing block is zero. Otherwise the last update gives $g \leq 2t$, proving the first branch.

For $1 < t \leq 2$, normalize the first row and column to be nonnegative by independent sign changes. A positive entry of the first Schur complement is at most one, so the second pivot is negative. Write the tail as

$$\begin{pmatrix} -t & \gamma_0 \\ \eta_0 & \zeta_0 \end{pmatrix}.$$

Let the relevant first-row and first-column coordinates be $x, u, y, v \in [0, 1]$, and set

$$c = t - 1, \quad s = xv, \quad z = uy, \quad r = uv, \quad w = xy.$$

The original entry at the pivot is $w - t$, whence $w \geq c$. Furthermore $sz = wr$ and $r \geq sz$. Intersecting the original entry bounds with the Schur pivot

bounds gives

$$\begin{aligned}\gamma_0 &\in [-1 - \min(c, s), 1 - s], \\ \eta_0 &\in [-1 - \min(c, z), 1 - z], \\ \zeta_0 &\in [-1 - \min(c, r), 1 - r].\end{aligned}\tag{5}$$

We claim that, with $M = t^2(3 - t)$,

$$t(1 - r) + (1 + \min(c, s))(1 + \min(c, z)) \leq M,\tag{6}$$

$$t(1 + \min(c, r)) + (1 + \min(c, s))(1 - z) \leq M.\tag{7}$$

For (6), replace r by sz . When $s, z \leq c$ the resulting function increases in either variable, since its corresponding derivative is $1 - cz$ or $1 - cs$; its maximum is at $s = z = c$. If $s \leq c \leq z$, it is $t[2 + s(1 - z)] \leq t[2 + c(1 - c)] = M$. The other mixed case is symmetric. If $s, z \geq c$, the bound is $t(1 - c^2) + t^2 = M$.

For (7), the shared-product relations imply $z \geq cr$, and also $z \geq r$ whenever $s \leq c$. The latter includes $s = 0$: then $sz = wr$ and $w \geq c > 0$ force $r = 0$. Thus no division by s is needed. Separate four closed cases. If $r \geq c, s \geq c$, then $z \geq cr \geq c^2$, and the expression is at most $t^2 + t(1 - c^2) = M$. If $r \geq c, s \leq c$, then $z \geq r \geq c$, and the bound $t^2 + t(1 - c) = 2t \leq M$ suffices. If $r \leq c, s \leq c$, the expression is at most

$$t(1 + r) + (1 + s)(1 - r) \leq 2t \leq M;$$

the first difference is $(c - s)(1 - r) \geq 0$, and $M - 2t = tc(1 - c) \geq 0$. Finally, if $r \leq c, s \geq c$, then $r - z \leq r(1 - c) \leq c(1 - c)$. The expression is at most $t[2 + c(1 - c)] = M$. Interchanging s, z gives the symmetric form of (7).

Put $A_s = 1 + \min(c, s)$, $B_s = 1 - s$, and similarly for z, r . By (5),

$$t\zeta_0 + \gamma_0\eta_0 \leq tB_r + \max(A_sA_z, B_sB_z) \leq M.$$

The first product uses (6); the second is at most $t + 1 \leq M$. On the other side,

$$-(t\zeta_0 + \gamma_0\eta_0) \leq tA_r + \max(A_sB_z, B_sA_z) \leq M$$

by (7) and its symmetric version. Thus the magnitude of the tail determinant is at most M , and division by t proves $g \leq t(3 - t)$.

Sharp examples are

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & t & t \\ 0 & -t & t \end{pmatrix} \quad (0 \leq t \leq 1), \quad \begin{pmatrix} 1 & 1 & c \\ 1 & -c & -1 \\ c & -1 & 1 \end{pmatrix}, \quad c = t - 1 \quad (1 \leq t \leq 2).$$

Finally $9/4 - t(3 - t) = (t - 3/2)^2$. $\square$

Corollary 3.2. *The first four pivot magnitudes of a normalized real completely pivoted matrix of order at most five satisfy*

$$p_1 = 1, \quad p_2 \leq 2, \quad p_3 \leq 9/4, \quad p_4 \leq 4.$$

Consequently $\rho(A, \pi) > \alpha$ implies $p_5 > \alpha$.

Proof. Any leading submatrix in the chosen pivot order is completely pivoted, and a Schur tail divided by its first pivot magnitude is normalized and completely pivoted. Put $x = p_2$, $y = p_3/p_2$ when these denominators are nonzero. Lemma 3.1 gives

$$xy \leq \Phi(x), \quad p_4 \leq x\Phi(y).$$

If $x \leq 1$, the second bound is at most $9/4$. If $1 \leq x \leq 3/2$, it is at most $27/8$. If $3/2 \leq x \leq 2$, then $y \leq 3 - x \leq 3/2$. The function Φ is nondecreasing on $[0, 3/2]$, so

$$p_4 \leq x\Phi(3 - x) = x^2(3 - x) \leq 4,$$

using $4 - x^2(3 - x) = (2 - x)^2(x + 1)$. Zero pivots imply zero subsequent growth. Equation (1) and $\alpha > 4$ finish the proof. $\square$

4. A COMPLETE TWO-STEP LIFT

All subsequent variables describe one common matrix. Let $u, x, v, q \in \mathbb{R}^3$, $p, k > 0$, and $e, \beta \in \mathbb{R}$. Set

$$M = \begin{pmatrix} 1 & -e & v^T \\ \beta & p - e\beta & (q + \beta v)^T \\ u & px - eu & D + xq^T + uv^T \end{pmatrix}, \quad D = kC, \quad C_{00} = 1. \quad (8)$$

Subscripts on 3×3 blocks run from zero to two. Its first two Schur tails are

$$M^{(2)} = \begin{pmatrix} p & q^T \\ px & D + xq^T \end{pmatrix}, \quad M^{(3)} = D. \quad (9)$$

Proposition 4.1. *Suppose C is completely pivoted in the fixed order, with $\|C\|_{\max} = 1$. Then (8) is normalized and completely pivoted, with first three pivot magnitudes $1, p, k$, if and only if*

$$|e|, |\beta|, \|u\|_{\infty}, \|v\|_{\infty}, \|x\|_{\infty} \leq 1, \quad (10)$$

$$|p - e\beta|, \|px - eu\|_{\infty}, \|q + \beta v\|_{\infty} \leq 1, \quad (11)$$

$$\|q\|_{\infty} \leq p, \quad \|D + xq^T\|_{\max} \leq p, \quad (12)$$

$$\|D + xq^T + uv^T\|_{\max} \leq 1. \quad (13)$$

Every normalized fixed-order completely pivoted matrix with three nonzero leading pivots has this representation after row-sign normalization of those pivots.

Proof. The identities (9) show sufficiency: the first matrix has entries bounded by one, the first tail by p , and the second tail is kC . Conversely recover e, β, u, v from the initial matrix, then p, x, q from its first Schur tail. The remaining tail is uniquely D ; division by its positive leading pivot gives C . The inequalities are precisely the initial and first-stage bounds. No independent completion is chosen for different inequalities. $\square$

The alternative notation $q = py$ makes the bottom block $kC + uv^T + pxy^T$. The actual variable q , rather than y , will be used in the seventeen-dimensional reduction. Write

$$C = \begin{pmatrix} 1 & a & b \\ c & ca + \rho & cb + \sigma \\ d & da + \tau & db + \omega \end{pmatrix}. \quad (14)$$

The fixed-order core conditions are $\|C\|_{\max} \leq 1$ and $|\sigma|, |\tau|, |\omega| \leq |\rho|$. For $\rho \neq 0$, set

$$h(C) = \omega - \tau\sigma/\rho, \quad F = k|h(C)|.$$

If $\rho = 0$, the final tail is zero; set $h(C) = F = 0$. Lemma 3.1 also yields

$$k \leq 9/4, \quad |h(C)| \leq 9/4, \quad k \leq 2p, \quad |h(C)| \leq 2|\rho|. \quad (15)$$

All these quantities belong to the same source.

5. CONSTRUCTIVE SATURATION OF THE LAST TAIL

The next reduction is an existence statement about representatives. It does not assert that every maximizer is already saturated.

Lemma 5.1 (Same-height diagonal saturation). *Every complete lift (8) with $F > 0$ has another complete lift with the same p, k, F whose normalized tail satisfies either*

$$D : \quad |\omega| = |\rho|, \quad |\sigma|, |\tau| \leq |\rho|,$$

or

$$X : \quad |\sigma| = |\tau| = |\rho|, \quad |\omega| \leq |\rho|.$$

The construction is valid on every legal tied pivot path.

Proof. For $0 \leq \lambda, \mu \leq 1$, let $R_\lambda = \text{diag}(1, \lambda, 1)$, $K_\mu = \text{diag}(1, \mu, 1)$. Replace

$$C \mapsto R_\lambda C K_\mu, \quad (u, x) \mapsto (R_\lambda u, R_\lambda x), \quad (v, q) \mapsto (K_\mu v, K_\mu q).$$

The new full matrix is $\text{diag}(I_2, R_\lambda) M \text{diag}(I_2, K_\mu)$. Its initial and first-tail entries are only contracted in magnitude, while the pivots $1, p, k$ do not change. The core tail transforms as

$$(\rho, \sigma, \tau, \omega) \mapsto (\lambda\mu\rho, \lambda\sigma, \mu\tau, \omega). \quad (16)$$

Since $F > 0$, $\rho \neq 0$. Put

$$S_0 = |\sigma/\rho|, \quad T_0 = |\tau/\rho|, \quad W_0 = |\omega/\rho|, \quad S_0, T_0, W_0 \in [0, 1].$$

If $W_0 \geq S_0 T_0$, choose $\lambda = \max(T_0, W_0)$, $\mu = W_0/\lambda$. If W_0 vanished, then $S_0 T_0 = 0$ and the original height would be zero. Thus the division is valid and $\lambda\mu > 0$. Now $\lambda \geq T_0$, $\mu \geq S_0$, and $\lambda\mu = W_0$, so the new completely pivoted tail lies on D . If $W_0 \leq S_0 T_0$, take $\lambda = T_0, \mu = S_0$. Positivity of the original height again excludes $\lambda\mu = 0$; the new tail lies on X . At the positive-product junction $W_0 = S_0 T_0 > 0$, the constructions agree. Finally,

$$\omega - \frac{(\mu\tau)(\lambda\sigma)}{\lambda\mu\rho} = \omega - \frac{\tau\sigma}{\rho}.$$

This preserves F and all the complete early-stage conditions. No exchange of unequal pivots is introduced. $\square$

Zero fifth pivots are already safe by Corollary 3.2. They need not have a zero fourth-stage tail: a nonzero rank-one 2×2 tail can have zero last pivot. In contrast, a zero fourth pivot under complete pivoting does force the entire current tail to vanish. The lemma requires neither a zero-target extension nor equality of the two full-matrix constructions at a zero-product junction.

Actual sign changes on the final rows and columns reduce every X -tail to

$$H = \begin{pmatrix} r & r \\ r & w \end{pmatrix}, \quad r > 0, \quad |w| \leq r, \quad F = r - w. \quad (17)$$

Here H is the *actual* fourth-stage tail, whose entries are k times the normalized core-tail entries. Indeed, for $\begin{pmatrix} \epsilon_{1r} & \epsilon_{2r} \\ \epsilon_{3r} & W \end{pmatrix}$, multiply rows by (ϵ_1, ϵ_3) and columns by $(1, \epsilon_1 \epsilon_2)$. This gives (17), with $w = \epsilon_1 \epsilon_2 \epsilon_3 W$, and preserves all absolute matrix bounds. Similarly every D -tail has a form

$$H = \begin{pmatrix} r & s \\ t & \chi r \end{pmatrix}, \quad r > 0, \quad 0 \leq s, t \leq r, \quad \chi \in \{-1, 1\}. \quad (18)$$

If $\chi = 1$, its height is $r - st/r \leq r \leq 4$; a zero arm also gives height $r \leq 4$. If $s = r$ or $t = r$, exchanging the corresponding tied final columns or rows and normalizing signs gives (17) at the same height. The sole additional domain is therefore

$$H = \begin{pmatrix} r & s \\ t & -r \end{pmatrix}, \quad 0 < s, t < r, \quad F = r + \frac{st}{r}. \quad (19)$$

We call this the proper negative diagonal domain.

6. THE COMPLETE CROSS-SATURATED BOUND

For (17), the actual core matrix is

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r + cB \\ dk & r + dA & w + dB \end{pmatrix}. \quad (20)$$

All of (10)–(13) and $\|D\|_{\max} \leq k$ are retained. No rank, cofactor-sign, or auxiliary entry condition is assumed.

Proposition 6.1 (Certified complete X bound). *Every real complete lift with (20), $p, k, r > 0$, and $|w| \leq r$, satisfies $F = r - w \leq \alpha$.*

The proposition is assembled from four certified components. Appendix A and Supplementary Material S4 identify the component certificates and their source domains.

Complete source domain	Certified bound
$0 < k \leq 2$	$F < 1653/400$
$k \geq 21/10$	$F < \gamma$
$2 < k \leq 21/10, r > k$	$F < \gamma$
$\gamma/2 \leq k \leq 21/10, r \leq k, F \geq \gamma$	$F \leq \alpha$

Assembly. The first three rows leave $2 < k < 21/10, r \leq k$. If $F < \gamma$, (3) suffices. Otherwise $\gamma \leq F \leq 2r \leq 2k$, hence $k \geq \gamma/2$, and the last row applies. The boundary $r = k$ lies in the last row; $k = 2, 21/10$ have closed coverage. Internal sign and transpose normalizations are actual matrix operations imposing no extra rank or cofactor hypothesis. $\square$

The small- k certificate has 268817 nodes, 134409 contradiction leaves, and no open leaf. The accepted two low- r roots have the following counts:

Root	Nodes	Contradiction leaves	α -safe leaves	Open
I	3450095	1725047	1	0
II	5920895	2873490	86958	0

The counts identify accepted objects, rather than replacing their acceptance rules and domain maps. An α -safe leaf can contain feasible points; it is not an infeasibility certificate at γ .

A canonical near-peak component establishes its exact height by finite localization, invertible slack coordinates, and the same algebraic root α . It consumes previously proved canonical-representative and row-scale reconstruction lemmas. Their original proofs and earlier exclusion trees remain explicit dependencies. The X assembly does not treat a canonical representative as an unproved global-entry assumption. Supplementary Materials S3 and S4 identify these modules, their exact verification records, and their roles in the complete bound.

7. EXACT REDUCTION OF THE NEGATIVE DIAGONAL DOMAIN

Replace (20) by

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & s + cB \\ dk & t + dA & -r + dB \end{pmatrix}. \quad (21)$$

We initially allow $0 \leq s, t \leq r$, so the safe boundary is included. A *complete source* means (8), (10)–(13), $\|D\|_{\max} \leq k$, and these tail conditions. The normalized first Schur tail is a real 4×4 completely pivoted matrix. Corollary 3.2 gives

$$F \leq 4p. \quad (22)$$

Hence $F > 4$ requires $p > 1$. The head bound implies $e\beta \geq p - 1 > 0$. A simultaneous sign flip of the second row and column gives $e, \beta > 0$. Simultaneously reversing all three remaining rows and columns leaves D fixed and reverses x , allowing $x_0 \geq 0$. Simultaneously reversing just the last

two rows and columns reverses A, B, c, d , leaves x_0 fixed, and leaves the last tail fixed, allowing $A \leq 0$. No sign is imposed on the other receivers or on c, d .

Fix the seventeen-coordinate common framework

$$\vartheta = (k, A, B, c, d, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2). \quad (23)$$

We maximize over the six remaining actual variables (β, p, e, r, s, t) , in the normalized range $p \geq 1$, $0 \leq e, \beta \leq 1$. Every high source has such a representative by the preceding argument. The constructions give feasible paths, not independent necessary relaxations.

7.1. The right-prefix interval. Define

$$I_\beta = [0, 1] \cap \bigcap_{j=0}^2 \{b : |q_j + bv_j| \leq 1\}.$$

If $v_j = 0$, this condition is $|q_j| \leq 1$; otherwise its endpoints are $(-1 - q_j)/v_j$ and $(1 - q_j)/v_j$, sorted in increasing order. An empty interval rules out the fibre. When it is nonempty, set $\bar{\beta} = \max I_\beta$.

Lemma 7.1. *For a complete source with $p \geq 1$, $0 \leq e, \beta \leq 1$, increasing β linearly to $\bar{\beta}$, while fixing the other variables, preserves complete feasibility and F .*

Proof. The right-prefix bands define the convex interval I_β . The head upper bound $p - e\beta \leq 1$ improves, and its lower bound is automatic because $p \geq 1$ and $e\beta \leq 1$. No other condition or height depends on β . $\square$

7.2. The attained p, e capacity. If $\bar{\beta} = 0$, the head forces $p = 1$, safe by (22). Suppose $\bar{\beta} > 0$. Form the head row $(a_0, b_0) = (1, \bar{\beta})$. For every $x_i \neq 0$, add the row

$$a_i = |x_i| > 0, \quad b_i = \text{sgn}(x_i)u_i.$$

When $x_i = 0$, its prefix bound is automatic. There are at most four rows, with $0 < a_i \leq 1$, $|b_i| \leq 1$. The complete remaining prefix system is

$$a_i p - b_i e \leq 1, \quad 0 \leq e \leq 1, \quad p \geq 1. \quad (24)$$

The reverse bound is automatic from $a_i p \geq 0$ and $b_i e \leq 1$. Define capacities for $b_i \geq 0$ and for $b_i > 0 > b_j$, respectively:

$$C_i = \frac{1 + b_i}{a_i}, \quad C_{ij} = \frac{b_i - b_j}{a_j b_i - a_i b_j}.$$

All denominators are positive. Put

$$P = \min(C_i, C_{ij}), \quad e_* = \max\left(0, \max_{b_i > 0} \frac{a_i P - 1}{b_i}\right). \quad (25)$$

Lemma 7.2. *The maximum p in (24) is P ; (P, e_*) is feasible. Furthermore $1 \leq P \leq 2$, and at most six capacities occur.*

Proof. Writing $\ell_i = 1 - a_i p + b_i e \geq 0$, feasibility implies

$$1 + b_i - a_i p = \ell_i + b_i(1 - e) \geq 0 \quad (b_i \geq 0),$$

and

$$b_i - b_j - (a_j b_i - a_i b_j)p = b_i \ell_j - b_j \ell_i \geq 0 \quad (b_i > 0 > b_j).$$

Thus every capacity is necessary. Conversely positive b_i give lower bounds for e , negative b_j give upper bounds, and zero b_i constrain only p . The C_i make all lower bounds at most one; the C_{ij} make each lower bound at most each upper bound. A negative row additionally requires a nonnegative upper bound, namely $p \leq 1/a_j$. Its pairing with the head implies this:

$$\frac{\bar{\beta} - b_j}{a_j \bar{\beta} - b_j} \leq \frac{1}{a_j},$$

since $a_j \leq 1$, $b_j < 0$. The head lower bound $(P - 1)/\bar{\beta}$ is nonnegative. Therefore (25) satisfies every row. Each capacity is at least one, and the head capacity is $1 + \bar{\beta} \leq 2$. If the numbers of positive, zero, and negative rows are m, z, n , then their number of capacities is $m + z + mn$, where $m \geq 1$, $m + z + n \leq 4$, and hence at most six. $\square$

After Lemma 7.1, move (p, e) along the segment to (P, e_*) . The prefix polygon is convex and p increases. The actual matrices

$$D, \quad S = D + xq^T, \quad O = D + xq^T + uv^T$$

stay fixed. Thus $|q_j| \leq p$, $|S_{ij}| \leq p$ improve, and all other bands remain valid. This second path preserves F ; it does not require e itself to vary monotonically.

7.3. The attained tail maximum. At $p = P$, form joint intervals for the entries of D :

$$\begin{aligned} \ell_{ij} &= \max(-k, -P - x_i q_j, -1 - x_i q_j - u_i v_j), \\ U_{ij} &= \min(k, P - x_i q_j, 1 - x_i q_j - u_i v_j). \end{aligned} \tag{26}$$

The five entries in the zeroth row or column of D are fixed. Check their intervals and $|q_j| \leq P$. Failure makes the fibre empty: decreasing p cannot repair a failed stage bound, while original-matrix and core bounds do not depend on p . For the other entries put

$$\begin{aligned} r_- &= \max(0, \ell_{11} - cA, dB - U_{22}), & R &= \min(U_{11} - cA, dB - \ell_{22}), \\ s_- &= \max(0, \ell_{12} - cB), & s_+ &= U_{12} - cB, \\ t_- &= \max(0, \ell_{21} - dA), & t_+ &= U_{21} - dA, \\ S_* &= \min(R, s_+), & T_* &= \min(R, t_+). \end{aligned}$$

Theorem 7.3 (Complete six-variable fibre maximum). *For a nonempty high-value fibre at fixed ϑ , the construction yields a complete source with*

$$(\beta, p, e, r, s, t) = (\bar{\beta}, P, e_*, R, S_*, T_*),$$

and

$$\mathcal{B}(\vartheta) = R + \frac{S_* T_*}{R} = \max_{\text{complete six-variable fibre}} F. \quad (27)$$

The tail fibre is nonempty exactly when the fixed checks pass and

$$R > 0, \quad R \geq r_-, \quad S_* \geq s_-, \quad T_* \geq t_-.$$

Every original source reaches the maximum by three complete feasible paths with nondecreasing height.

Proof. The first two paths have been established. At their endpoint the tail feasible set is exactly

$$r_- \leq r \leq R, \quad s_- \leq s \leq s_+, \quad t_- \leq t \leq t_+, \quad r > 0, \quad 0 \leq s, t \leq r.$$

The intervals are computed simultaneously from the same u, x, v, q . Nonemptiness implies $R \geq r > 0$, $S_* \geq s$, $T_* \geq t$; then (R, S_*, T_*) satisfies every interval. The stated conditions are therefore necessary and sufficient. For any feasible tail,

$$Rr(\mathcal{B} - F) = (R - r)(Rr - st) + rt(S_* - s) + rS_*(T_* - t) \geq 0. \quad (28)$$

The factors have the required signs because $R \geq r$ and $st \leq r^2$. The segment to this maximum stays in the convex feasible set, with all three coordinates nondecreasing. Along it,

$$\partial_r F = 1 - st/r^2 \geq 0, \quad \partial_s F = t/r \geq 0, \quad \partial_t F = s/r \geq 0.$$

This proves the last path and the attained maximum. $\square$

Every hypothetical $F > \alpha$ source has its framework in the closed box

$$\begin{aligned} \frac{4\gamma}{9} \leq k \leq \frac{9}{4}, \quad -\frac{9}{4} \leq A \leq 0, \quad -\frac{9}{4} \leq B \leq \frac{9}{4}, \quad |c|, |d| \leq 1, \\ |u_i|, |v_i| \leq 1, \quad 0 \leq x_0 \leq 1, \quad |x_1|, |x_2| \leq 1, \quad |q_i| \leq 2. \end{aligned} \quad (29)$$

The lower bound on k follows from $F \leq (9/4)k$, and the other bounds from the complete bands and actual sign normalizations. The root retains the complete fibre conditions, not merely the box. In particular it does not discard the proper negative domain with $k \leq 2$. An additional shared head budget $pk \leq 4$, when used, follows from the leading 3×3 determinant: its magnitude is pk , and the maximum determinant on the real unit-entry cube is 4. For completeness, multilinearity reduces that maximum to sign matrices; their 3×3 determinants are multiples of four, while the Hadamard bound is $3\sqrt{3} < 8$.

All intersections, minima, and maxima include ties. Zero coefficients are handled before division. The point $R = 0$ cannot support a high source and is rejected as an empty positive-tail fibre. No early generic-position restriction is needed.

8. RATIONAL CERTIFICATES AND MIXED-HEIGHT COVERAGE

The remaining finite proof obligation on (29) is that every complete framework has $\mathcal{B}(\vartheta) \leq \alpha$. Discovery proposes subdivisions and terminal certificates. Acceptance checks them by exact rational arithmetic on their complete declared sources. Floating-point optimization may guide discovery but is not an acceptance criterion.

8.1. Necessary outer systems. Each actual source gives all coordinates and products simultaneously. Necessary polynomial inequalities may be relaxed over a rational box, provided the relaxation contains this same source. For example, for $L_x \leq x \leq U_x$, $L_y \leq y \leq U_y$, the shared product $z = xy$ satisfies

$$\begin{aligned} z &\geq L_x y + L_y x - L_x L_y, & z &\geq U_x y + U_y x - U_x U_y, \\ z &\leq L_x y + U_y x - L_x U_y, & z &\leq U_x y + L_y x - U_x L_y. \end{aligned}$$

A product appearing in several rows is not assigned independent physical values. Enclosures are outward. Sign-sensitive division or use of a selected min/max formula requires qualification on the entire box, or subdivision that retains every alternative.

If the necessary rows are $a_j \cdot z \leq b_j$, exact nonnegative weights give

$$a = \sum_j \lambda_j a_j, \quad B = \sum_j \lambda_j b_j, \quad a \cdot z \leq B, \quad \lambda_j \geq 0.$$

A box lower bound for $a \cdot z$ greater than B rules out the source. The acceptor checks all weights, row identities, endpoints, and strict comparisons as rationals. An interval contradiction is the one-expression case. The actual height can be introduced as a shared coordinate through

$$rF - r^2 - st = 0,$$

using both inequality directions. It must not be replaced by the X -specific objective $r - w$.

8.2. Height certificates and feasible transports. Suppose $F = h \cdot z$, after introducing a height coordinate if necessary. For $t > 0$, let $c = a - th$ and

$$m = \sum_i \min(c_i L_i, c_i U_i).$$

Then

$$tF = a \cdot z - c \cdot z \leq B - m. \tag{30}$$

Thus $B - m \leq t\alpha_-$ proves $F \leq \alpha$. It does not establish infeasibility.

Other safe terminals use feasible transformations to a domain already bounded by α . If every source z in a box has a complete image $T(z)$ with $F(T(z)) \geq F(z)$, and $F(T(z)) \leq \alpha$, that box is safe. The transformation must preserve the actual coupled matrix bands, rather than changing unrelated abstract capacities.

A uniformly certified family near cross-saturated tails illustrates this rule. Introduce nonnegative gaps $\sigma = r - s, \tau = r - t$ in

$$H = \begin{pmatrix} r & r - \sigma \\ r - \tau & w \end{pmatrix}, \quad F = \frac{(r - \sigma)(r - \tau)}{r} - w.$$

At fixed p , twenty actual band slacks and $(r + w, \sigma, \tau)$ form twenty-three coordinates. For each of four rational centres c_j , an exact preconditioner C_j and the full Jacobian J satisfy, throughout $\|z - c_j\|_\infty \leq 1/1250$,

$$\|I - C_j J\|_\infty < 1/25.$$

Whole-box certificates also bound the two inverse-coordinate directions $J^{-1}e_\sigma, J^{-1}e_\tau$ in infinity norm by 3, their height derivatives by $-1/10$, and all unselected physical slacks strictly away from zero. Hence a source satisfying

$$\|z - c_j\|_\infty + 3(\sigma + \tau) < 1/1250$$

can follow the negative σ , then negative τ , direction to a complete X source inside the same certified box. The selected slacks and $p, r + w$ remain fixed. Uniform inverse-Jacobian and strict distance bounds ensure existence to the endpoint. Integration gives

$$F(z) \leq \alpha - \frac{\sigma + \tau}{10}. \quad (31)$$

The rational centres and matrices are archived proof data; no numerical trajectory establishes the assertion. In particular there is no exchange of unequal pivots before reaching a tied boundary. Further safe transformations obey the same source and height-direction requirements.

8.3. Complete composition.

Lemma 8.1 (Mixed threshold and algebraic coverage). *Let $\gamma < \alpha$. Let a rational root Q_0 contain a canonical image of every hypothetical source with $F > \alpha$, with height at least that of the source. Suppose a finite acyclic rooted proof graph covers Q_0 by exact subdivisions, qualified contractions, and source-preserving coordinate changes that do not decrease the height of a hypothetical source. Suppose every terminal establishes either:*

- (1) *no complete source with $F \geq \gamma$ exists in its region; or*
- (2) *every complete source in its region has $F \leq \alpha$.*

If every child is accounted for and each terminal source is bound to its declared root path, then no original source has $F > \alpha$.

Proof. A hypothetical source has a canonical image in Q_0 with no smaller height, still exceeding $\alpha > \gamma$. At a subdivision it belongs to a closed child; overlaps on boundaries are harmless. A qualified contraction has already certified its discarded region impossible or safe. A coordinate change maps the same source into its declared image at no smaller height. Following these relations through the finite complete graph reaches a terminal and contradicts either terminal assertion. $\square$

Previously accepted subproofs may be composed by this lemma, but only on the same mathematical sources. A matching dimension or a cached success label is insufficient. The implementation binds model, parent, path, endpoints, transformation prefix, rule implementation, and proof payload. The final traversal reconstructs the original root boxes and verifies every grafted subproof’s source. Hashes establish byte identity; mathematical validity comes from the accepted subproof and its declared rule.

9. THE COMPLETE NEGATIVE-DIAGONAL CERTIFICATE

Proposition 9.1 (Certified complete negative-diagonal bound). *Every complete real source with (21), $p, k, r > 0$, and $0 < s, t < r$, satisfies*

$$F = r + st/r \leq \alpha.$$

The computation uses the full seventeen-variable root (29) with Theorem 7.3’s exact framework semantics. Its model contains every hypothetical $F > \alpha$ source after the three actual paths; it imposes no selected tail-contact equation on all canonical images. Supplementary Materials S5–S7 specify the model, acceptance rules, and complete certificate data.

9.1. Base certificate and remaining source domains. Exact rational verification of the base certificate accepted 749693 nodes: 374846 splits, 374839 contradiction terminals, four safe terminals, and four open terminals. Its 750 enclosed continuation certificates had no internal open terminal. The four remaining domains are closed dyadic boxes reconstructed from the original seventeen-dimensional root. Denote them by Q_1, Q_2, Q_3, Q_4 , in the order specified in Supplementary Material S7. The base certificate proves the required bound outside their union.

Complete certificates for Q_1, Q_3, Q_4 cover, respectively, 111, 159, and 56 constituent sources. The certificate for Q_2 has 240 constituent sources; 239 are covered by its component certificates and the remaining one by the local completion described below. Every constituent source retains its complete ancestor transformation prefix.

9.2. Local completion. The remaining source in Q_2 is covered by a binary subtree with 54 distinct, prefix-free terminal paths. Their union exhausts that subtree. The terminal justifications are

Terminal justification	Count
Contradiction certificates	48
Height and feasible-transport certificates	3
Canonical-source threshold exclusions	3
Total	54

Each canonical-source threshold exclusion covers both qualified charts of its declared source. The three height or transport certificates establish safety at α . The mathematical verifiers and source-prefix checks were executed for

this local completion. It covers one constituent source of Q_2 ; full coverage of Q_2 also uses the other 239 sources.

9.3. Composition at the original root. The local completion is attached to its exact source in Q_2 , preserving the already verified complement. The four complete certificates are then attached to the four open terminals of the base certificate. The composition traverses the actual base-root structure and reconstructs its entire set of open boxes. For each attachment it checks the original model, root path, box endpoints, and full transformation prefix against the corresponding complete certificate. The reconstructed boxes coincide with Q_1, Q_2, Q_3, Q_4 . Thus every open base-root source is covered by a matching complete proof, and the assembled certificate has no uncovered source.

The original-parent coverage count is 469/469. This count provides a consistency check; coverage follows from the complete root traversal and exact source comparisons. Supplementary Material S7 records the paths, endpoints, certificate identities, and composition results needed to check these bindings.

Proof of Proposition 9.1. A hypothetical $F > \alpha$ source has $p > 1$ by (22). Its actual sign normalization and Theorem 7.3's paths produce a complete framework in (29) with $\mathcal{B} > \alpha$. The accepted base proof and the four full-parent completions form the finite complete cover just described. Each contradiction terminal excludes this high source, and each safe terminal bounds its height by α . All original root paths and transformed sources are accounted for. Lemma 8.1 therefore excludes the hypothetical source. $\square$

Remark 9.2 (Verification and composition). The final composer reuses identified, previously accepted complete-root and parent receipts. It verifies their producer and payload identities and exact source bindings; it does not rerun every terminal's mathematics. The base tree retains four open records, each discharged by an attached complete certificate. The records retain their original meaning in this composition. Likewise a source safe at α need not be infeasible at the lower threshold γ .

10. COMPLETION OF THE GLOBAL PROOF

Proof of Theorem 1.2. The lower bound is [2, Theorem 2.2]; Section 2 describes the exact corresponding witness in the archive.

For the upper bound, fix an arbitrary nonzero real matrix and arbitrary legal complete-pivoting path. Normalize its maximum initial entry to one and absorb the path's permutations into the matrix. If a pivot vanishes, its whole trailing block vanishes. Corollary 3.2 bounds all growth through the fourth stage by $4 < \alpha$. It remains to exclude a fifth-pivot value larger than α .

Proposition 4.1 gives the complete two-step lift. Lemma 5.1 gives a same-height complete source on the D or X saturated boundary. Actual sign

changes put every X source into (17), bounded by Proposition 6.1. For a D source, (18) with positive sign or a zero arm has height at most $r \leq 4$; a tied arm goes to X . Every remaining source belongs to (19), so Proposition 9.1 applies. No fifth pivot exceeds α . Since the matrix and legal path were arbitrary, $\rho_5^{\mathbb{R}} \leq \alpha$, proving equality. $\square$

11. REPRODUCIBILITY AND SCOPE

The computational proof separates discovery, verification, and composition. Discovery constructs cuts, subdivisions, and feasible transformations. Verification checks finite rational inequalities and qualified analytic rules on complete sources. Composition establishes that verified pieces cover the original domain with matching source data. This separation allows component proofs to be checked and reused independently.

The supplementary documentation is organized into eight modules, listed in Appendix A. It identifies the polynomial and algebraic witness, analytic reductions, complete X certificates, negative-diagonal model, terminal rules, and final coverage graph. The accompanying source and dependency index specifies exact file identities, source domains, verification entry points, and the scope of each recorded run. Large certificate payloads are separate from the readable documentation; full verification requires those payloads and all referenced component dependencies.

The base-root verification used exact rational arithmetic and took 22815.35 recorded seconds. The final local verification took approximately 184.37 seconds. These are the times for their respective verification scopes, not the total cost of discovery or of checking the entire proof. Later parent and root compositions reuse the identified component verification records while checking exact source bindings. A fresh end-to-end verification therefore requires execution of every necessary component verifier as well as the final composition. Checking a file’s hash alone establishes identity, not mathematical validity.

A companion Lean development formalizes the algebraic constant and its attainment, key analytic reductions, and a conditional final theorem. The latter retains two explicit global safety hypotheses, for the physical X domain and the original B -root capacity endpoints. The present proof addresses these obligations through the written analytic arguments, exact certificates, and source-composition logic; their complete discharge in Lean, including the remaining model and coverage connections, is not claimed. The Python/C++ verifier implementations are not formally verified. This is a computer-assisted proof with partial Lean formalization. Component verifications and final composition are distinguished in Supplementary Material S8; composition does not re-execute every earlier verifier.

Data and code availability. The supplementary documentation accompanies the manuscript and identifies the code, models, and certificate payloads used

in the proof. Requests for the full computational materials may be directed to qianli.ma@zju.edu.cn.

The theorem determines the maximum value and supplies an attaining matrix. Classification of all maximizers, uniqueness modulo symmetry, and the analogous complex-matrix value remain separate questions.

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APPENDIX A. PROOF MODULES AND SUPPLEMENTARY MATERIAL

Supplementary Materials S1–S8 organize the analytic and computational dependencies by their mathematical role. The theorem statements in the text specify the domains on which each component is used.

Supplementary unit	Mathematical content and role
S1: Constant and attainment	The degree-61 polynomial, exact isolating interval, identity with the constant of [2], and supplementary algebraic reconstruction of an attaining matrix. The lower bound is [2, Theorem 2.2].
S2: Scalar bounds and complete coordinates	The conditional three-pivot envelope, the first-four-pivot bound, the complete two-step lift, and the same-height D/X saturation proved in Sections 3–5. The saturation statement used here requires positive fifth-pivot height.
S3: Canonical representative and height	Qualified representative existence, exact row-scale reconstruction, and canonical localization and height certificates, established independently of the final negative-diagonal bound. Six complete exclusion trees contain 228542 nodes and 114274 contradiction leaves. The subsequent 152467-record localization uses 208274 contraction certificates; its three terminal boxes are covered by the algebraic height theorem.
S4: Complete cross-saturated domain	The four source domains and exact sign and transpose maps of Proposition 6.1, including their shared boundaries. The low- r component uses S3. All statements concern complete real matrices; no rank, cofactor-sign, or auxiliary entry assumption is added.

Supplementary unit	Mathematical content and role
S5: Negative-diagonal framework	The three complete feasible paths, attained six-variable maximum, and seventeen-variable root of Section 7. Zero factors, capacity ties, and the proper negative-diagonal range $k \leq 2$ are retained.
S6: Exact acceptance rules	Rational interval and nonnegative-combination certificates, qualified contractions, algebraic-height rules, and complete feasible transports. Contradictions above γ and safety at α have the distinct semantics of Lemma 8.1.
S7: Complete negative-diagonal cover	The accepted base proof, complete parent proofs, and exact source bindings of Section 9. The final 54-terminal source completes the last parent’s 240/240 responsibilities and the total 469/469 original parents. Four source-bound completions cover the base tree’s four retained open records; no original source region remains uncovered.
S8: Reproduction and dependency index	Model and payload identities, acceptance records, executable dependencies, and source-composition maps. The final composition’s 357 direct dependencies are supplemented by their recursive dependencies; the direct list alone is not the complete proof archive.

The dependency direction is S3 to S4, and then S4 to the cross-saturated safety rules used in S6–S7; the final negative-diagonal cover is not an input to S3 or S4. Counts identify accepted objects, while their rules and exact source bindings establish validity and coverage. An α -safe region may contain feasible matrices.

Component acceptance and final composition are separate operations. The latter reuses identified acceptance records and checks their source bindings; it does not rerun every component’s mathematics. Accordingly, the completed composition is not claimed to be a new from-scratch replay of the entire dependency chain. The supplementary index distinguishes the recorded acceptance runs, the checks performed during composition, and the materials required for independent reproduction.

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